Monte Carlo and Importance Sampling

Approximating Intractable Integrals Stochastically

Edward Milsom

2026-10-06

Overview

Slide 1 Recap — Bayesian Marginalisation Scenarios

Given observed data $\mathbf{y}$ and assumed model $\mathcal{M}$ with variables $\boldsymbol{\theta}$, there are three main scenarios where we want to marginalise or integrate out variables:

- Making predictions for a new data point y_*:

$$p\left(y_*|\mathbf{y},\mathcal{M}\right) = \int p\left(y_*|\boldsymbol{\theta},\mathcal{M}\right) p\left(\boldsymbol{\theta}|\mathbf{y},\mathcal{M}\right) d\boldsymbol{\theta}$$

- Identifying/quantifying specific parameter uncertainty:

$$p\left(\theta_1|\mathbf{y},\mathcal{M}\right) = \int p\left(\theta_1,\theta_2|\mathbf{y},\mathcal{M}\right) d\theta_2$$

- Computing evidence for model $\mathcal{M}$:

$$p\left(\mathbf{y}|\mathcal{M}\right) = \int p\left(\mathbf{y}|\boldsymbol{\theta},\mathcal{M}\right) p\left(\boldsymbol{\theta}|\mathcal{M}\right) d\boldsymbol{\theta}$$

Slide 2 Recap — Bayesian Marginalisation Scenarios

Inconvenient Truth The majority of these integrals will be analytically intractable.

In most cases, we can't find closed-form solutions for them.

Lectures 8 & 9 focus on approximating these integrals using stochastic techniques.

We'll look at *deterministic* techniques next semester.

Slide 3 In this Episode ...

- Computing expectations
- General principle of "Monte Carlo" sampling
- Sampling in a single dimension:
    - Inverse-CDF
- The "Curse of Dimensionality" (with demo)
- Sampling in low dimensions:
    - Importance sampling
- Sampling in higher dimensions:
    - A first introduction to **Markov Chain Monte Carlo** (MCMC)

Expectations & Monte Carlo Sampling

Slide 4 Computing Expectations

- Many (Bayesian) integrals of interest can be considered special cases of the more general problem of computing expectations of a given function over a given probability distribution:

$$F = \int f(\theta)\, p(\theta)\, d\theta = \langle f(\theta) \rangle_{p(\theta)}$$

- *e.g.* a predictive distribution:

$$\begin{aligned} p(y_*|\mathbf{y}) &= \int p(y_*|\boldsymbol{\theta})\, p(\boldsymbol{\theta}|\mathbf{y})\, d\boldsymbol{\theta} \\ &= \left\langle p(y_*|\boldsymbol{\theta}) \right\rangle_{p(\boldsymbol{\theta}|\mathbf{y})} \end{aligned}$$

- When the expectation (integral) cannot be evaluated analytically (annoyingly often!), then we need to find some workable approximation

- This lecture introduces stochastic Monte Carlo approximation methods

---

Slide 5 The "Monte Carlo" Sampling Principle

- We approximate the expectation:

$$F = \langle f(\boldsymbol{\theta}) \rangle_{p(\boldsymbol{\theta})} = \int f(\boldsymbol{\theta})\, p(\boldsymbol{\theta})\, d\boldsymbol{\theta}$$

- By the random, Monte Carlo, sample:

$$\widehat{F} = \frac{1}{R} \sum_{r=1}^{R} f\left(\boldsymbol{\theta}^{(r)}\right)$$

- Where $\boldsymbol{\theta}^{(r)}$ is an independent random sample from $p(\boldsymbol{\theta})$

- It may be shown that as $R \to \infty$, then $\widehat{F} \to F$

Slide 6 Monte Carlo Sampling — Expected Accuracy

- How accurate is this approximation for realistic R?
- The expected error (variance) of $\widehat{F}$ is:

$$\epsilon^2 = \int \left(\widehat{F} - F\right)^2 p(\boldsymbol{\theta})\, d\boldsymbol{\theta}$$

- If σ_f^2 is the variance of f:

$$\sigma_f^2 = \int \left\{ (f(\boldsymbol{\theta}) - F)^2 \right\} p(\boldsymbol{\theta})\, d\boldsymbol{\theta}$$

- ... then as R increases, ϵ^2 decreases proportional to σ_f^2/R

Key result Accuracy of $\widehat{F}$ does not depend on the dimensionality of $\boldsymbol{\theta}$

Slide 7 Monte Carlo Sampling in Practice

- The problem now becomes one of obtaining samples $\boldsymbol{\theta}^{(r)}$ from $p(\boldsymbol{\theta})$
- This is likely to be non-trivial!
- Potential problems:

1. we know $p(\boldsymbol{\theta})$, but can't sample from it directly (*i.e.* no available *Python* library functions)
2. $p(\boldsymbol{\theta})$ is not in a standard form, or is un-normalised

- There is a method for addressing in the case of certain *univariate* distributions
- Problem is universal in Bayesian inference, and will be our main focus

Single dimension: Inverse CDF Sampling

Slide 8 The Inverse-CDF Method

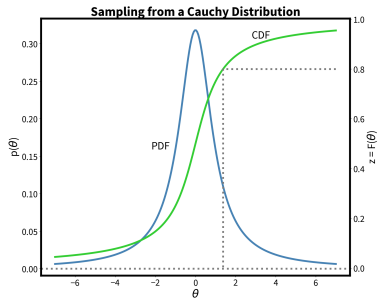
- We would like to sample from $p(\theta)$, but cannot do so directly
- Assume we know:
  - its cumulative density function (c.d.f.):
$$F(\theta) = P(\theta \leq x) = \int_{-\infty}^{x} p(\theta)\, d\theta$$
  - and its inverse c.d.f. $F_\theta^{-1}(z)$, with $z \in [0, 1]$
- Then, we sample z uniformly from $[0, 1]$ and compute the transformation $y = F_\theta^{-1}(z)$
- From which we may verify:
$$p(y) = p(z)\left|\frac{dz}{dy}\right| = p(\theta)$$

Slide 9 The Inverse-CDF Method — Cauchy Example

Its inverse c.d.f.:
$$F_\theta^{-1}(z) = \tan\left[\pi\left(z - 1/2\right)\right]$$

The Cauchy distribution:

$$p(\theta) = \frac{1}{\pi}\frac{1}{\theta^2 + 1}$$



Increasing dimension: Typical Scenarios

Slide 10 Typical Scenario: Un-normalised Distributions

- Consider that the distribution to be sampled from is written as:

$$p(\mathbf{\theta}) = \frac{P^*(\mathbf{\theta})}{Z}$$

- The numerator $P^*(\mathbf{\theta})$ specifies the form (shape) of the distribution
- The denominator Z is the normalising constant (scaling factor):

$$Z = \int P^*(\mathbf{\theta})\, d\mathbf{\theta}$$

- Very often, we know $P^*(\mathbf{\theta})$, but not Z

Fortunately … There are several sampling methods that only require $P^*(\mathbf{\theta})$ in order to approximate expectations over $p(\mathbf{\theta})$.

Slide 11 Example of Identifying $P^*(\mathbf{\theta})$: Bayesian Linear Regression

- Parameter posterior (over weights $\mathbf{w}$):

$$p(\mathbf{w}|\mathbf{y}, \alpha, \sigma^2) = \frac{\text{likelihood}\times\text{prior}}{\text{normalising factor}} = \frac{p(\mathbf{y}|\mathbf{w}, \sigma^2)\, p(\mathbf{w}|\alpha)}{p(\mathbf{y}|\alpha, \sigma^2)}$$

- The posterior $p\left(\mathbf{w}|\mathbf{y},\alpha,\sigma^2\right)$ and normalising factor (marginal likelihood) $p\left(\mathbf{y}|\alpha,\sigma^2\right)$ are both analytically computable

- But extending to hyper-parameters α and σ^2:

$$p\left(\alpha,\sigma^2|\mathbf{y}\right) = \frac{p\left(\mathbf{y}|\alpha,\sigma^2\right)\ p\left(\alpha\right)\ p\left(\sigma^2\right)}{p\left(\mathbf{y}\right)}$$

- The posterior $p\left(\alpha,\sigma^2|\mathbf{y}\right)$ is not computable since there is no closed-form solution for the normalisation integral $p\left(\mathbf{y}\right)$

- However, we can define $P^*\left(\alpha,\sigma^2\right) = p\left(\mathbf{y}|\alpha,\sigma^2\right)\ p\left(\alpha\right)\ p\left(\sigma^2\right)$

Slide 12 Sampling in High Dimensions — Caution!

- In the last lecture, we estimated a good value for basis radius r by approximating $p\left(\mathbf{y}|\alpha,\sigma^2,r\right)$ using a Monte Carlo sample evaluated over a regular grid of $\log\alpha$ and $\log\sigma$ values

- **Caution!** This approach works fine in low dimensions (computation aside) ...

- But ... to obtain the benefit of $1/R$ reduction in our estimate's error, we need representative samples $\boldsymbol{\theta}^{(r)}$ from $p\left(\boldsymbol{\theta}\right)$

- In particular, samples from all areas of significant probability mass

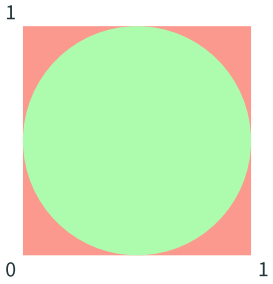
- But how do we find those regions in high dimensions?

- Exhaustive grid search is impractical, even for well-behaved distributions, due to the infamous curse of dimensionality

The “Curse of Dimensionality”

Slide 13 The “Curse of Dimensionality”

- Our intuitions often break down in higher dimensions
- Consider sampling uniformly from a circular p.d.f. in two dimensions:



- In 2D, the green area represents 78.5% of the data space
- But this volume gets vanishingly small as dimension increases

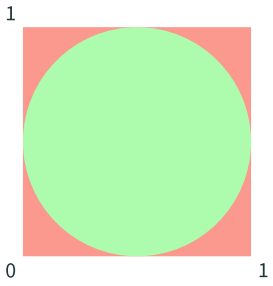
---

Slide 14 The “Curse of Dimensionality”

How much volume is in the corners?

For a D-dimensional hypersphere, what proportion of the parameter space lies in the corners of the hypercube?

D	Sphere Volume	Corner Proportion
2	0.785	21.5%
3	0.524	47.6%
4	0.308	69.2%



D	Sphere Volume	Corner Proportion
5	0.164	83.6%
…	…	…
10	0.002	99.8% !!!

Slide 15 **Next …**

There are two principal methods which attempt to *localise* the areas of high probability in $P^*(\boldsymbol{\theta})$:

1. Rejection sampling (not covered here)
2. Importance sampling

   Both approaches introduce the concept of a proposal distribution that is "close" to $P^*(\boldsymbol{\theta})$ in some sense

Low dimension: Importance Sampling

Slide 16 **Importance Sampling — Overview**

- Not a method to sample from $p(\theta)$, but to compute expectations of arbitrary functions w.r.t. $p(\theta)$:

$$\langle f(\theta) \rangle_{p(\theta)} = \int f(\theta)\, p(\theta)\, d\theta$$

- Again, assume $p(\boldsymbol{\theta}) = P^*(\boldsymbol{\theta})/Z$ and we can compute $P^*(\boldsymbol{\theta})$

- We then define a proposal distribution, ideally close to $p(\theta)$

$$Q(\theta) = Q^*(\boldsymbol{\theta})/Z_Q$$

- We need to be able to:
  - generate samples from $Q(\theta)$
  - evaluate $Q^*(\theta)$

Slide 17 Importance Sampling — Procedure

- Generate R samples $\theta^{(r)}$ from $Q(\theta)$
- Define importance weights:

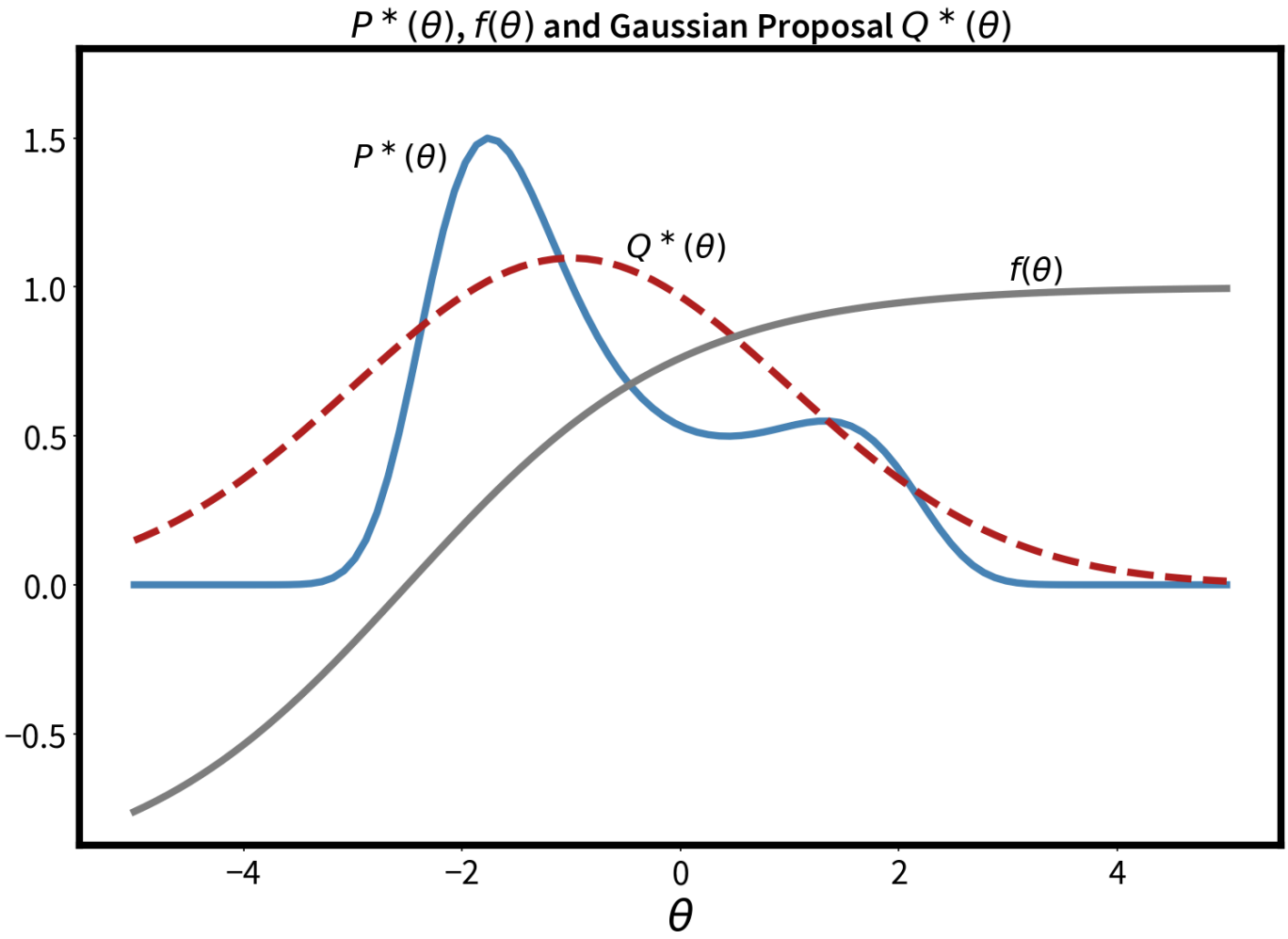
$$w_r = \frac{P^*\left(\theta^{(r)}\right)}{Q^*\left(\theta^{(r)}\right)}$$

- Compute weighted Monte Carlo expectation:

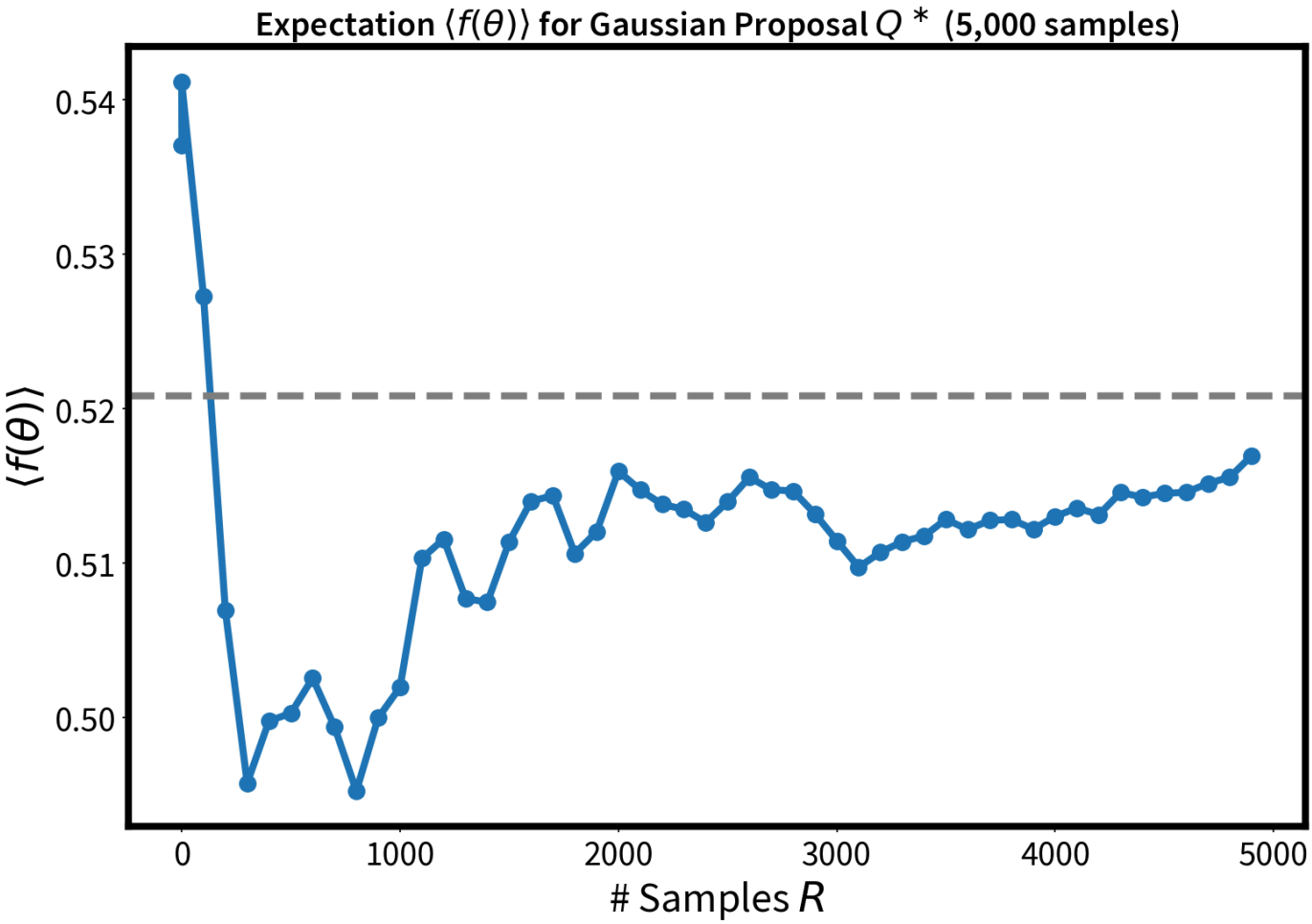
$$\widehat{F} = \frac{1}{\sum_r w_r} \sum_{r=1}^{R} w_r f\left(\theta^{(r)}\right)$$

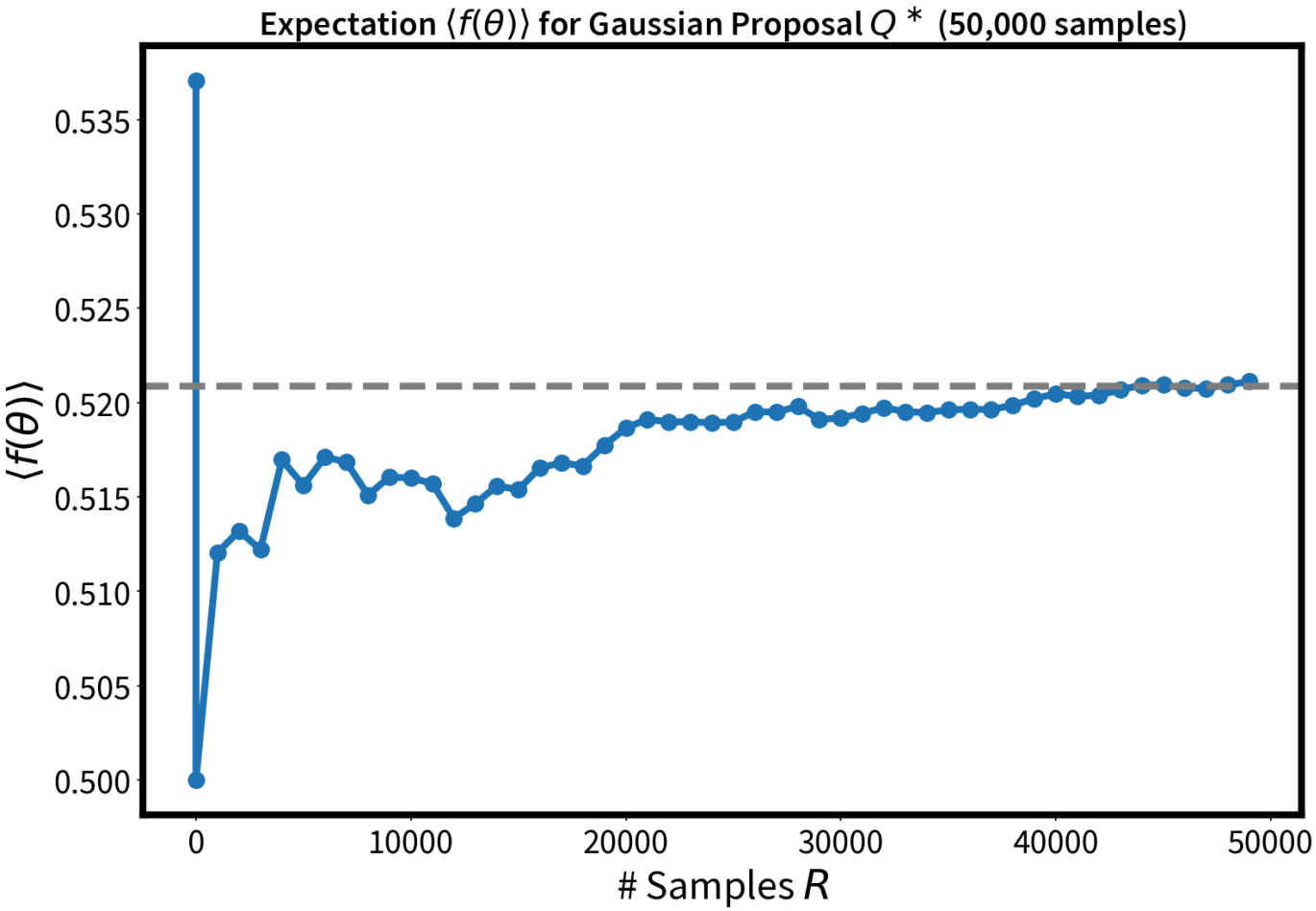
- Done!

Slide 18 Importance Sampling — Illustration

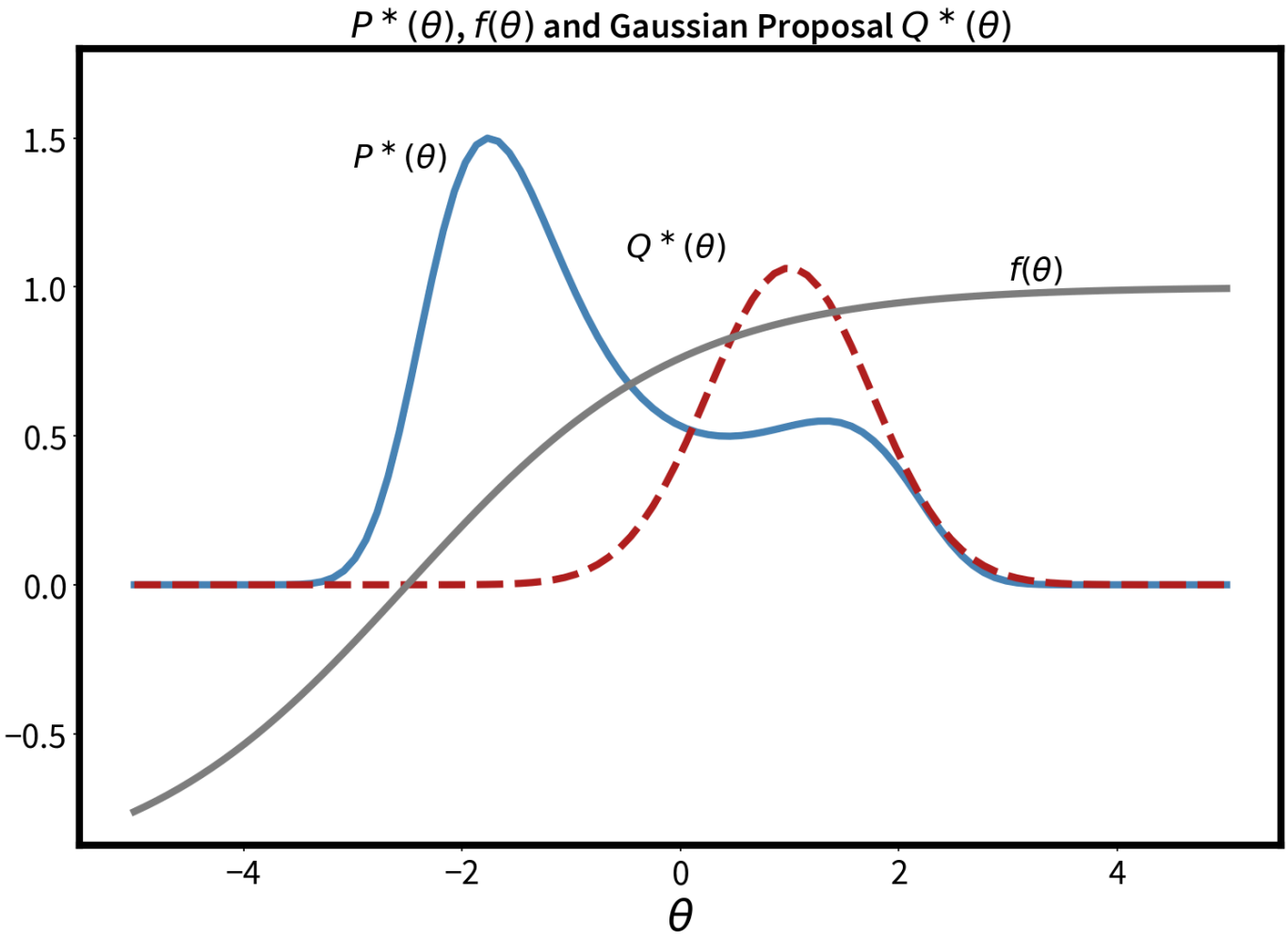


Slide 19 Importance Sampling — Expectation Estimate

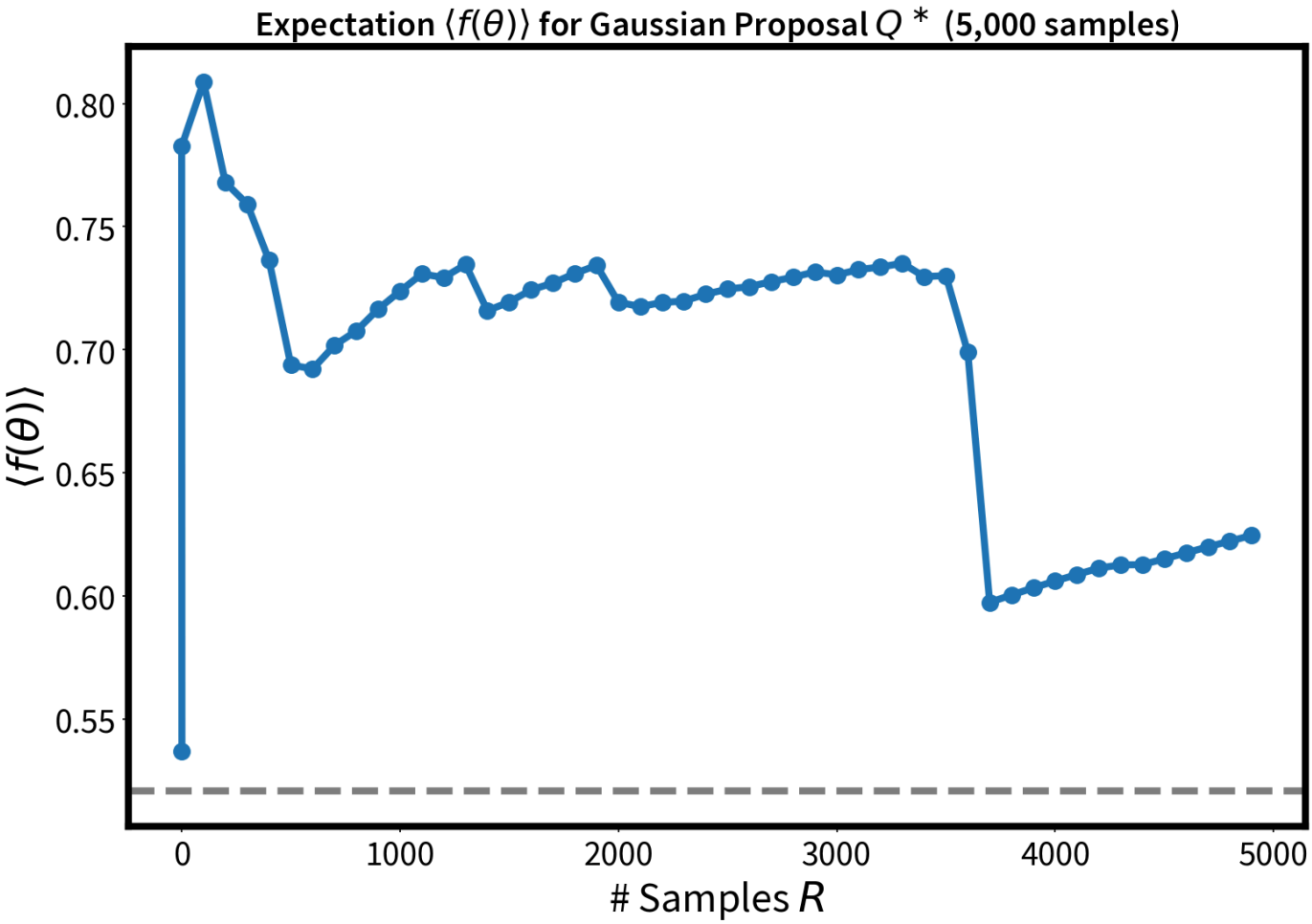


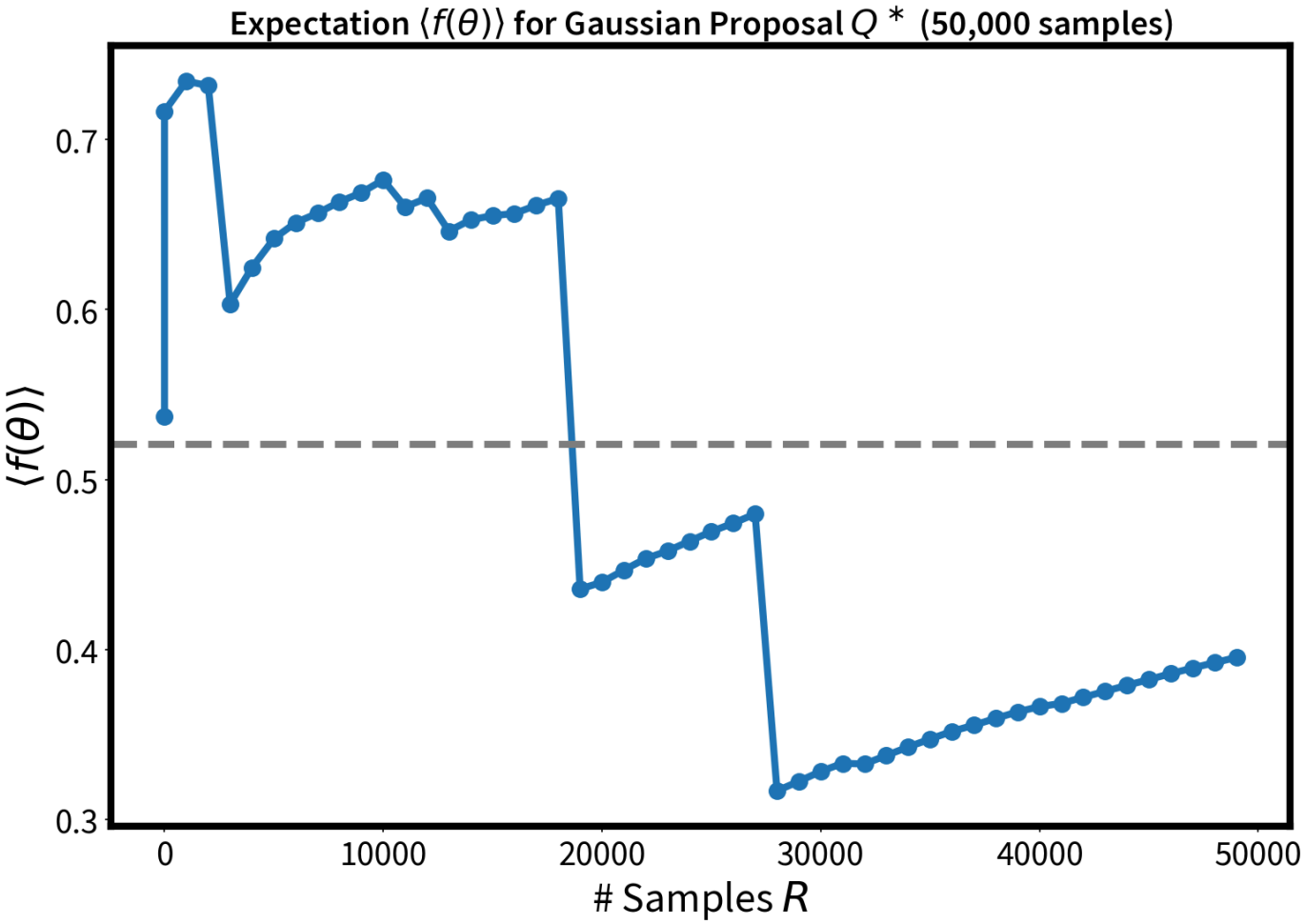

Expectation ⟨f(θ)⟩ for Gaussian Proposal Q * (50,000 samples)
0.535
0.530
0.525
0.520
0.515
0.510
0.505
0.500
⟨f(θ)⟩
0
10000
20000
30000
40000
50000
Samples R

Slide 20 Importance Sampling — A Bad Proposal!

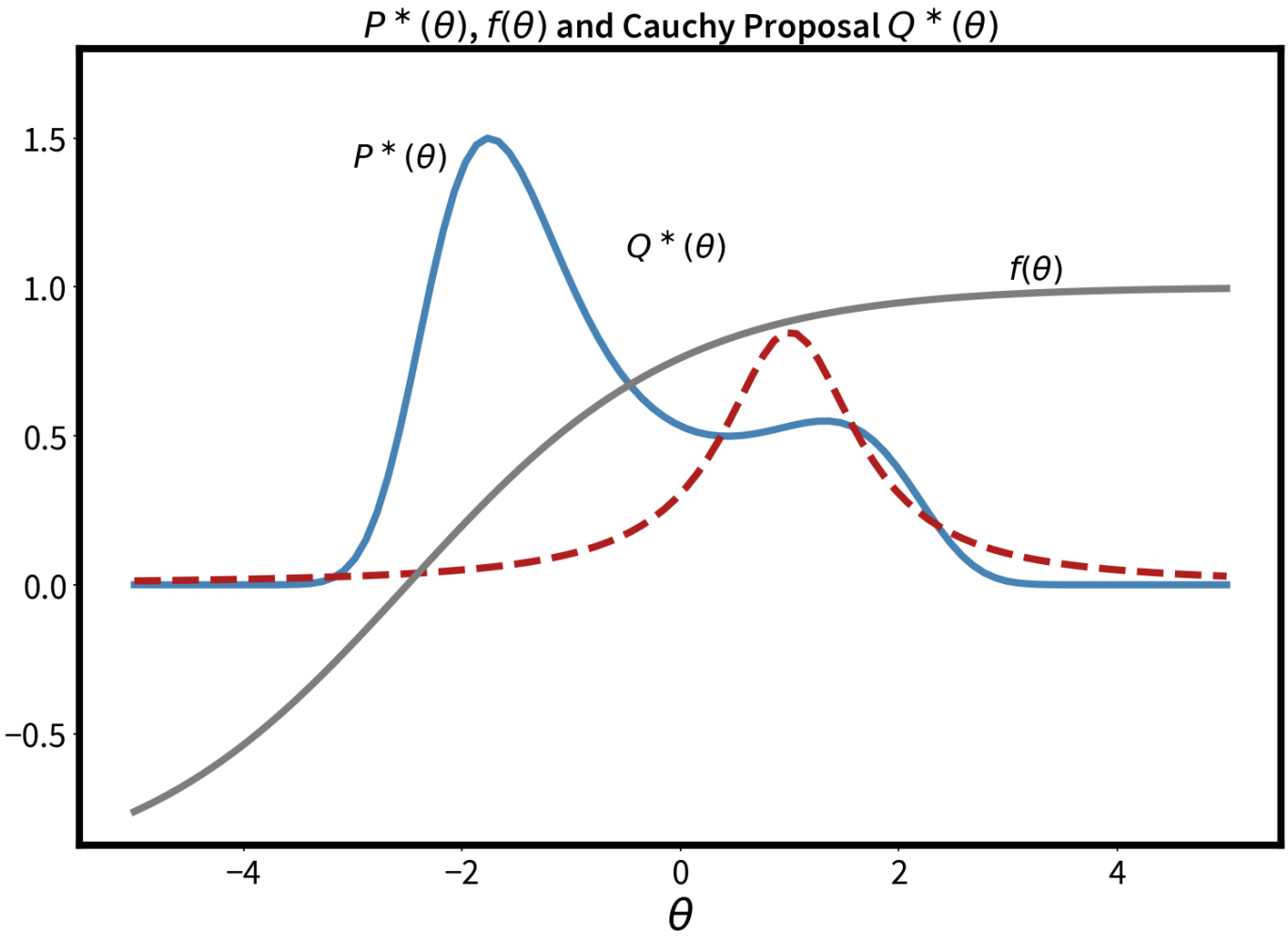


Slide 21 Importance Sampling — Non-converging Estimate

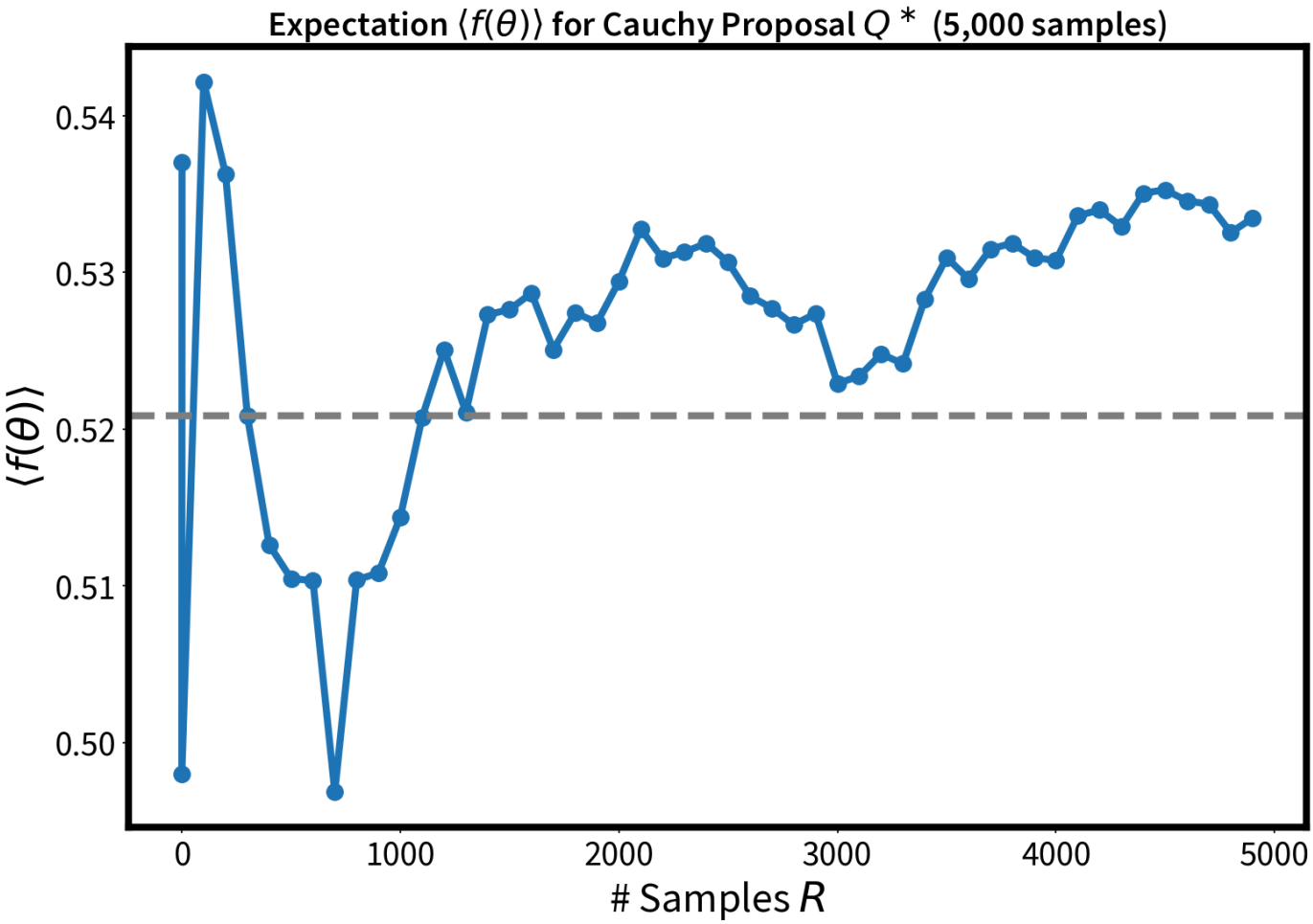


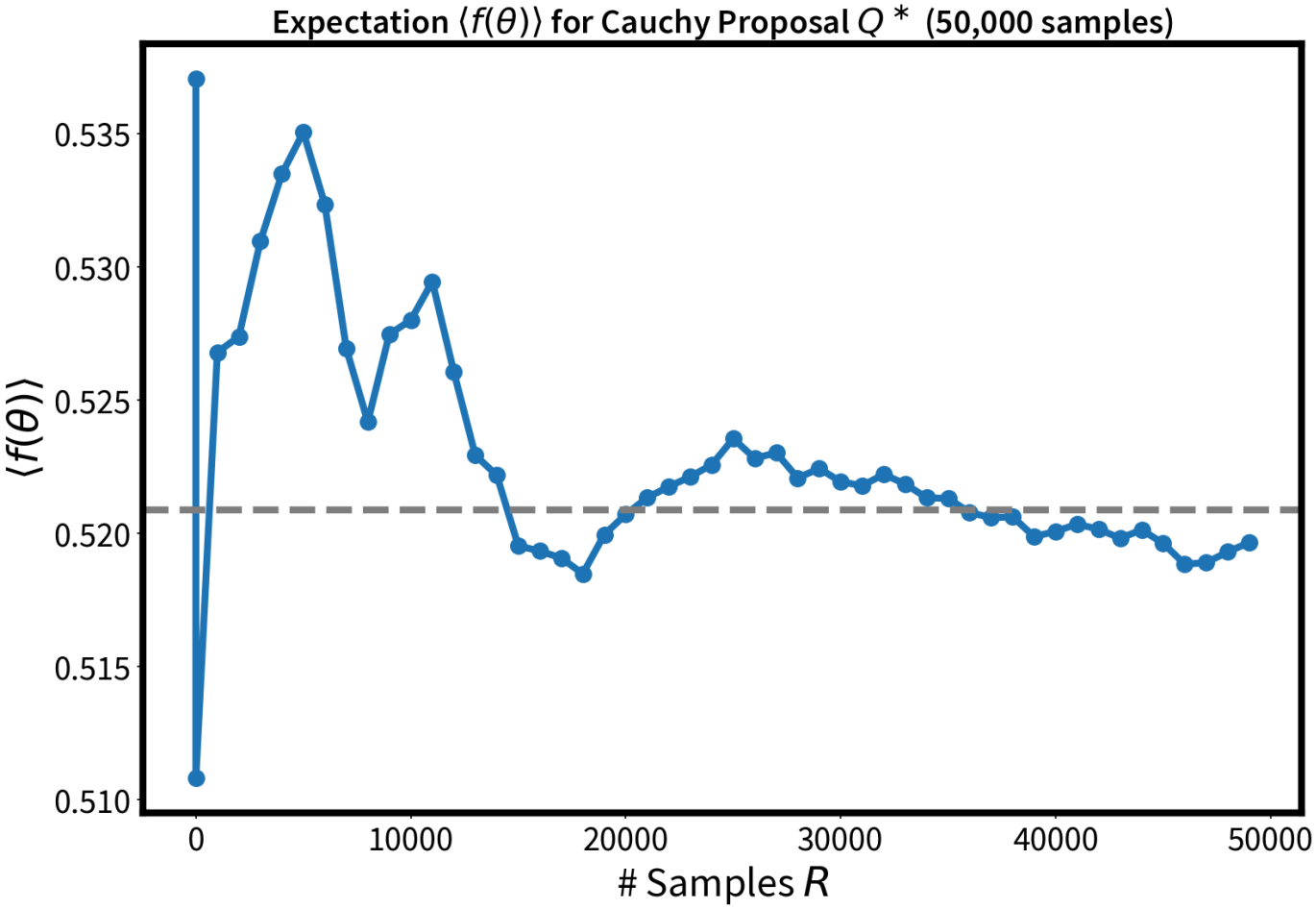

Expectation ⟨f(θ)⟩ for Gaussian Proposal Q * (50,000 samples)
⟨f(θ)⟩
0.7
0.6
0.5
0.4
0.3
0
10000
20000
30000
40000
50000
Samples R

Slide 22 Importance Sampling — Cauchy Proposal Distribution



Slide 23 Importance Sampling — Better Estimate





Slide 24 **Importance Sampling — Summary**

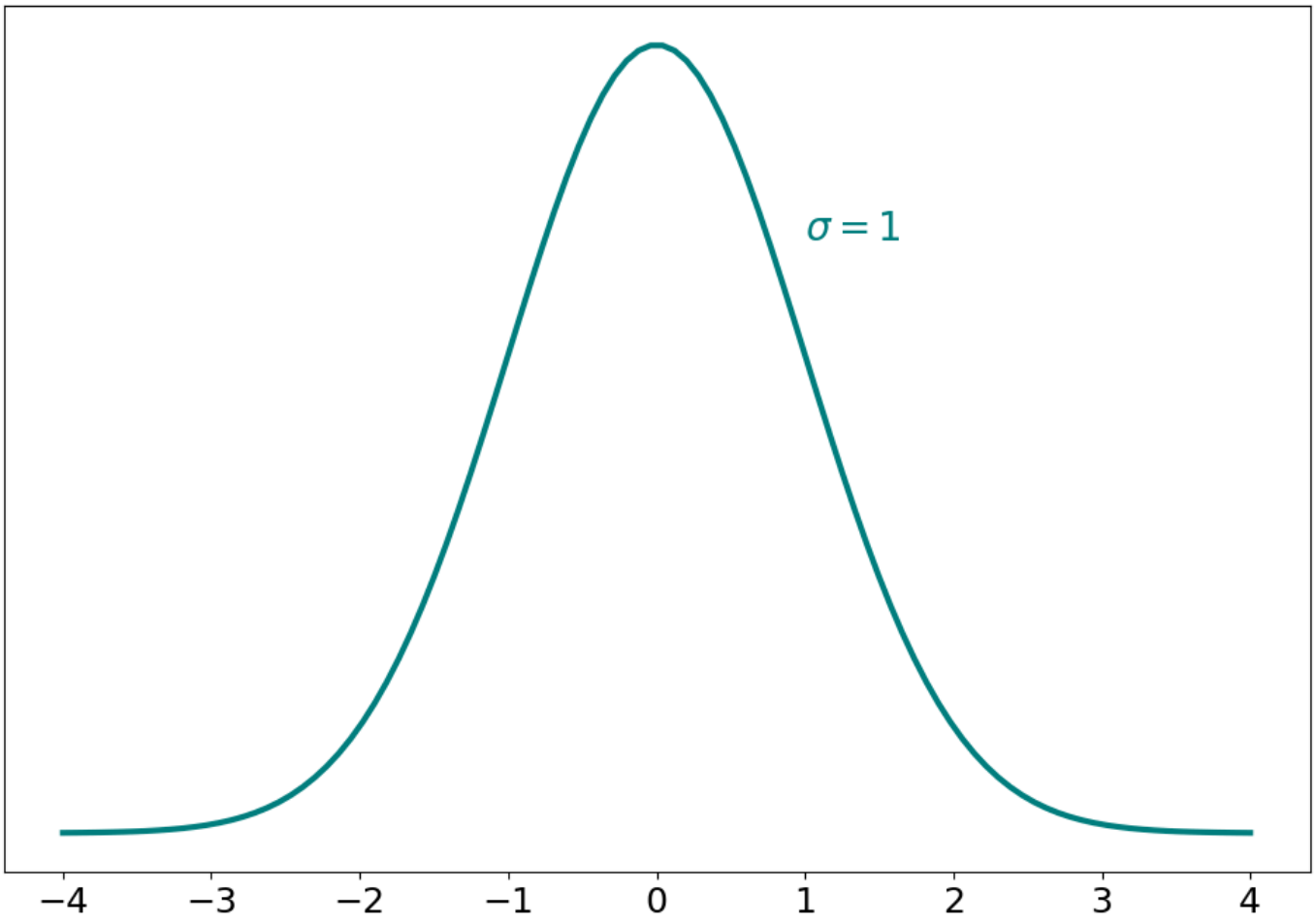
- Efficiency and accuracy require that $Q(\theta)$ is a good approximation to $P(\theta)$
- A good $Q(\theta)$ is hard to find
  - In particular, $Q(\theta)$ should not be small where $P(\theta)$ has significant probability mass (the “bad proposal”)
- Generally, $Q(\theta)$ should be heavy-tailed
- Conclusion: importance sampling is not practical in high dimensions

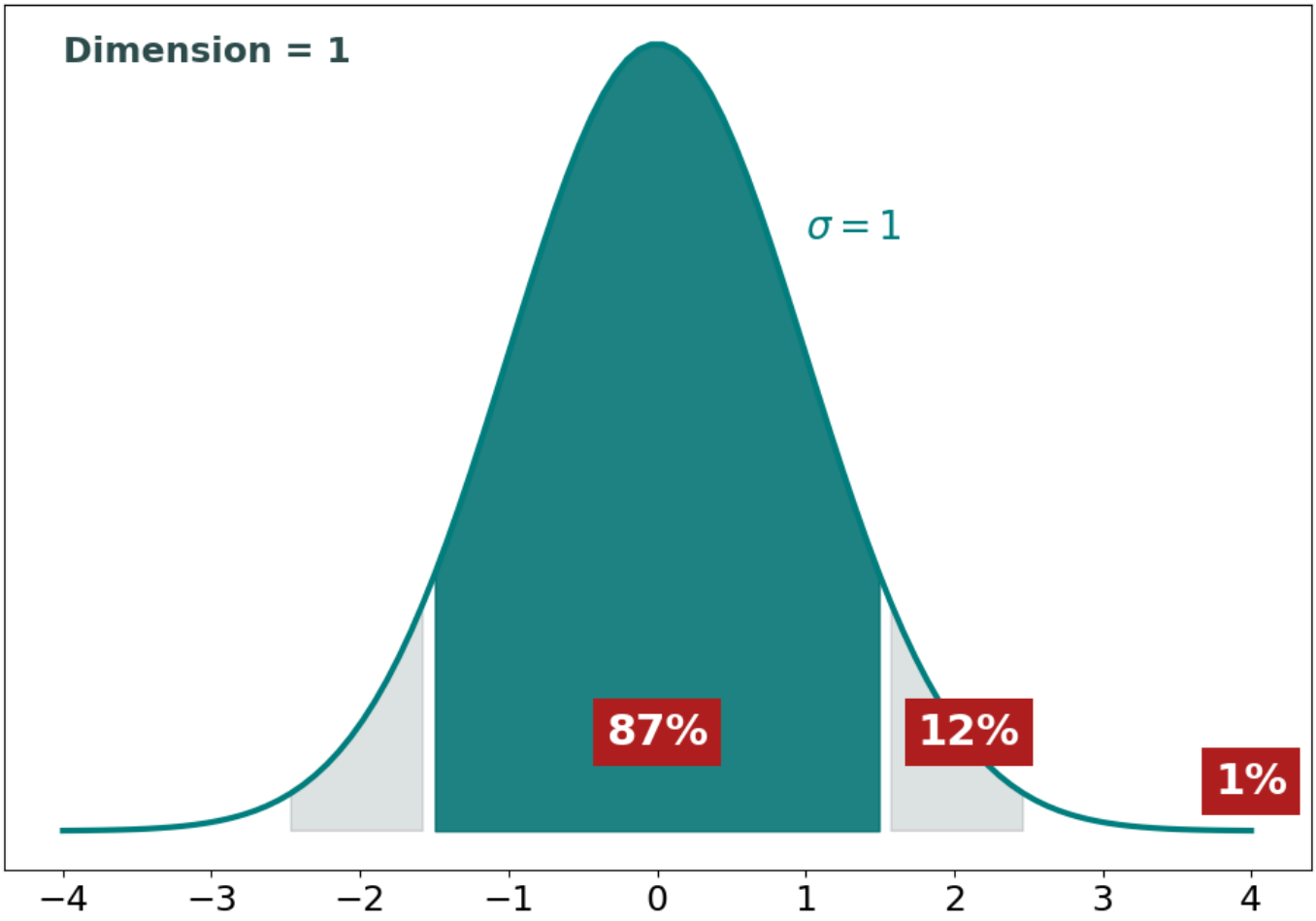
- though it still has application in certain lower-dimensional settings

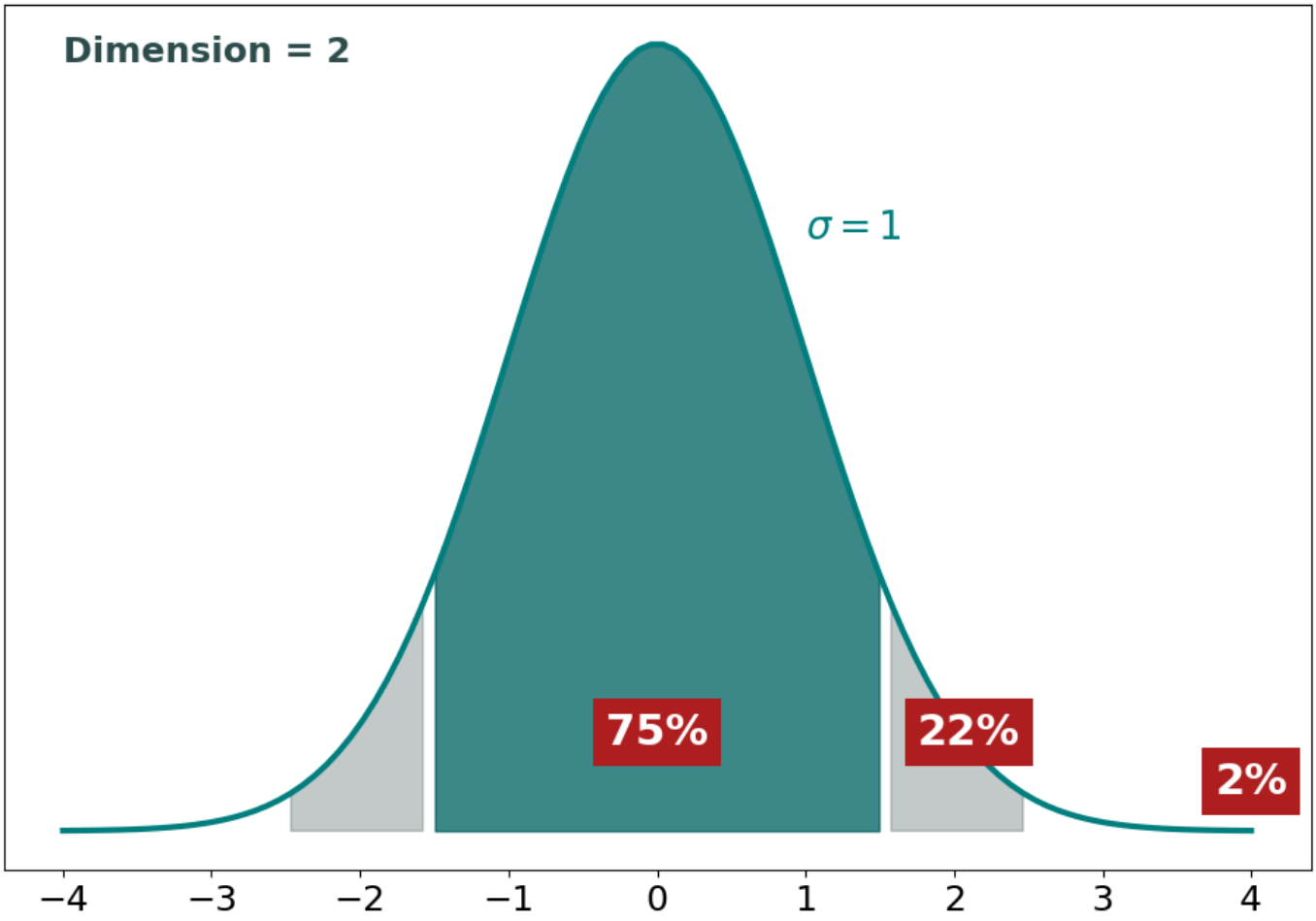
- For higher dimensions, we need: Markov Chain Monte Carlo

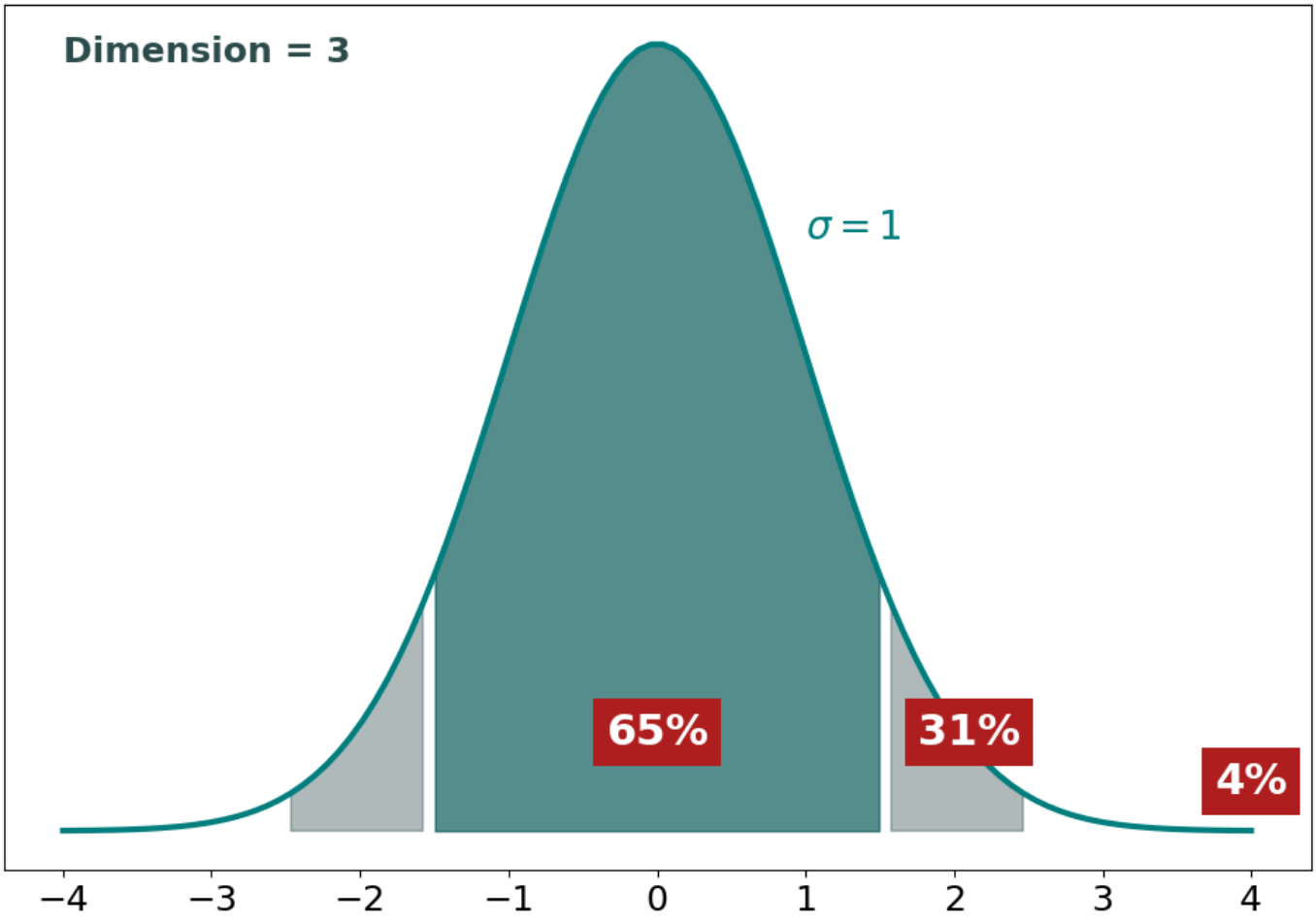
Slide 25 Further Motivation: The "Curse of Dimensionality" Again!

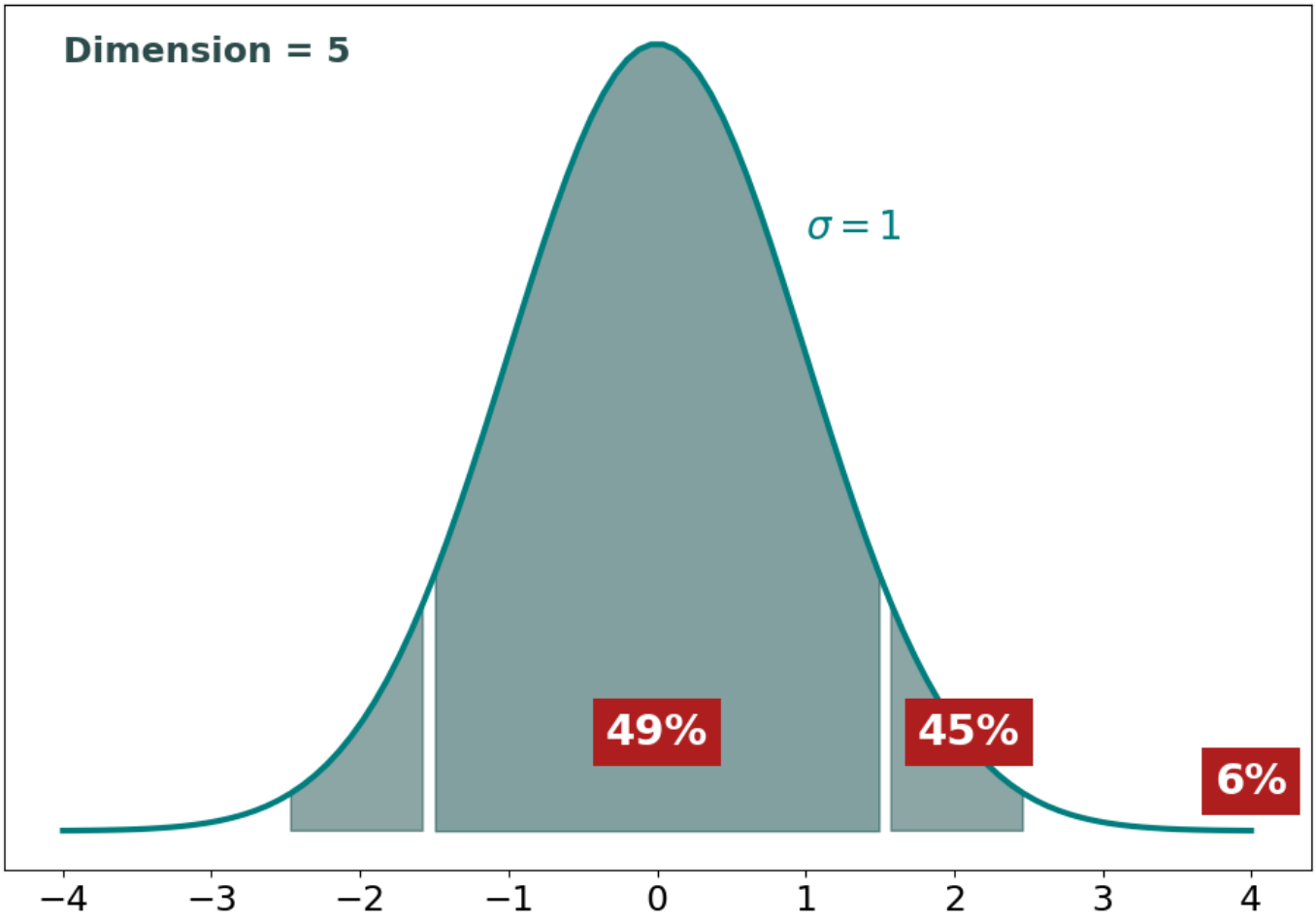
Where is the Gaussian probability mass?

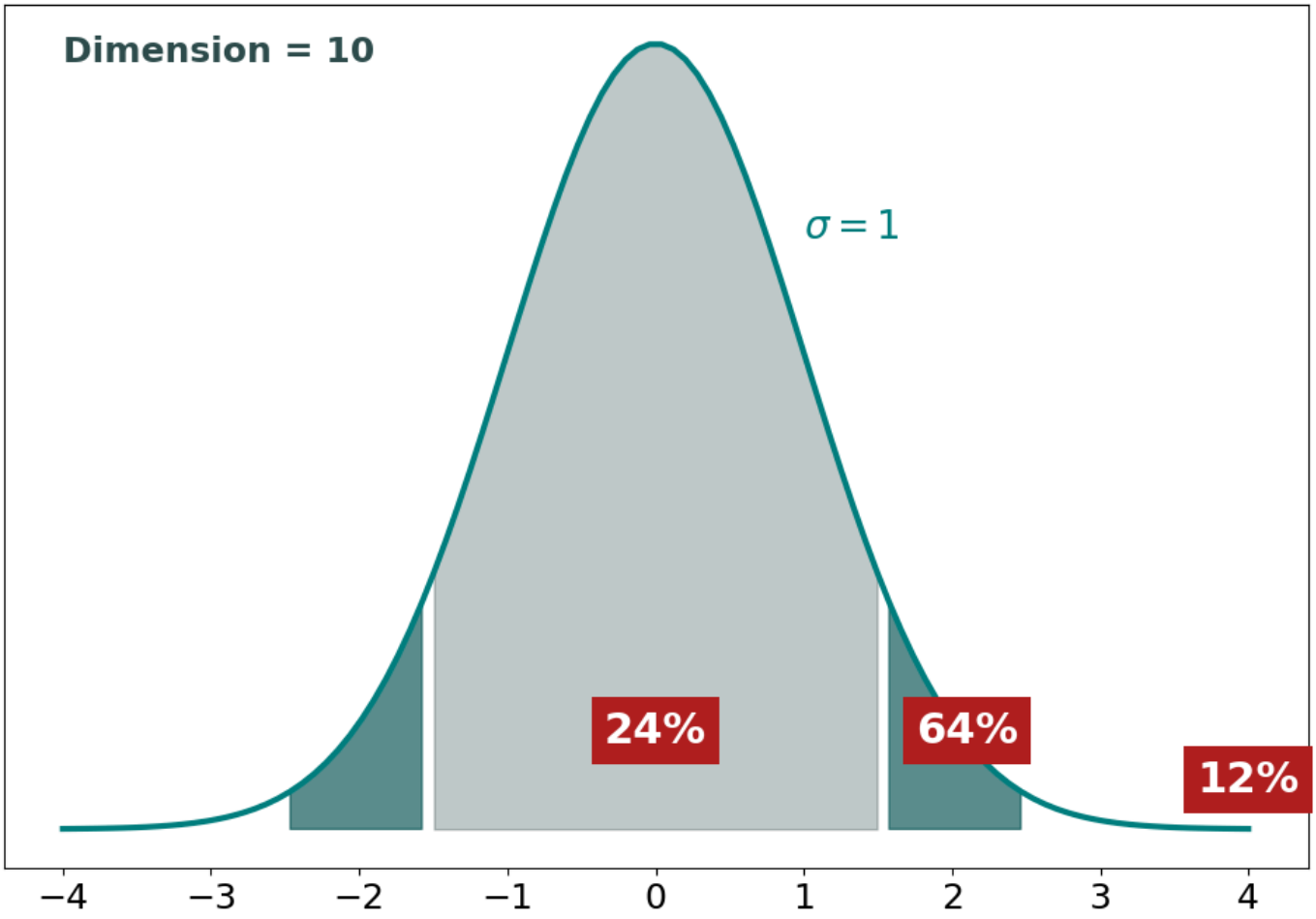


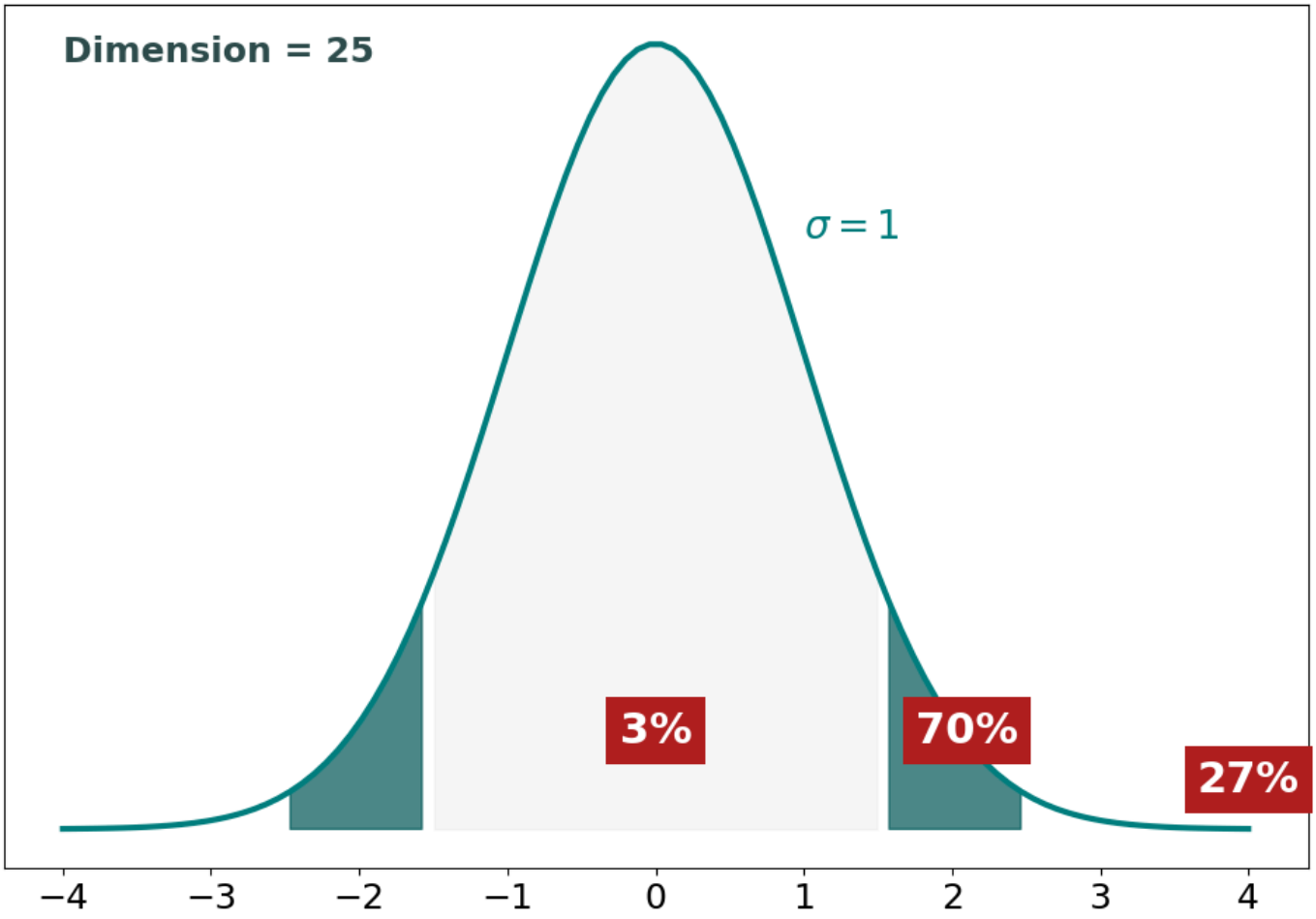

Dimension = 1
$\sigma = 1$
87%
12%
1%
−4
−3
−2
−1
0
1
2
3
4


Dimension = 2
$\sigma = 1$
75%
22%
2%
−4
−3
−2
−1
0
1
2
3
4


Dimension = 3
$\sigma = 1$
65%
31%
4%
−4
−3
−2
−1
0
1
2
3
4


Dimension = 5
$\sigma = 1$
49%
45%
6%
−4
−3
−2
−1
0
1
2
3
4


Dimension = 10
$\sigma = 1$
24%
64%
12%
−4
−3
−2
−1
0
1
2
3
4



High dimension: Markov Chain Monte Carlo

Slide 26 Markov Chain Monte Carlo (MCMC) — Preview

- It is difficult to find a good proposal distribution $Q(\boldsymbol{\theta})$ that is close enough to $P(\boldsymbol{\theta})$ to be practical
- Especially in high dimensions!
- Markov Chain Monte Carlo (MCMC) is a sampling method designed to obtain samples from $P(\boldsymbol{\theta})$ without need for a "globally close" $Q(\boldsymbol{\theta})$

- Instead, it makes use of a "local" proposal $Q\left(\boldsymbol{\theta}^{'}|\boldsymbol{\theta}^{(t)}\right)$ that in principle allows it to "track" regions of high probability mass

- We'll look at this next week ...